# Mushfique Shahriar Aurnab

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## Education

**Daegu Gyeongbuk Institute of Science and Technology (DGIST)** — Daegu, Republic of Korea
*B.S. student; intended focus in Computer Science and Electrical Engineering* — *Aug. 2024 – Present*

**Maastricht University, Center for European Studies** — Maastricht, Netherlands
*Summer Programme in European Studies* — *Summer 2026*

- Coursework: Global Governance of AI; Intercultural Communication.

## Experience

**Undergraduate Researcher, UGRP I** — DGIST
*Comparative Study of Catastrophic Forgetting in CNNs and Vision Transformers* — *Spring 2026*

- Designed a nine-run sequential CIFAR-10 diagnostic across three class splits and three seeds, using linear probes at six depths to distinguish classifier-readout failure from deeper representation loss.
- At Task-B epoch 25, 79.7% of misclassified old-task examples remained linearly recoverable; refitting the full linear readout restored 74.6% mean old-task accuracy.
- Compared five replay-selection strategies and retained the negative result: random replay outperformed collision-, fate-, memorization-, and speed-based targeting in buffer-only recovery.

**Computer Vision Lab, DGIST** — Daegu, Republic of Korea
*Student Research Assistant – Video Dataset Annotation* — *May 2025 – Aug. 2025*

- Prepared video dataset annotations under Prof. Sunghoon Im for experiments in video–text alignment and text-based object detection.

## Personal Projects

**GPX Motion Analysis** | *Python, time-series alignment, Matplotlib* — 2026

- Rebuilt a motion-data project as a tested CLI that aligns irregular GPX tracks on a shared grid while preserving segment boundaries and long gaps as missing evidence.
- Exports aligned samples, motion metrics, threshold sensitivity, summary JSON, and figures from deterministic synthetic fixtures; CI checks Python 3.11 and 3.14.

**Vox Pop** | *Python, MCP, FastEmbed* — 2026

- Built a Python library, CLI, and MCP server for searching nine public discussion platforms; a semantic router selects relevant communities before retrieval.

**Course Extractor** | *Python, Requests, BeautifulSoup* — 2026

- Implemented FOSE endpoint discovery with JSON-LD and HTML fallbacks; validated structured JSON and Markdown output on 508 Harvard DCE courses with zero reported page errors.

**MapleStory Code Battle AI** | *Python, OpenCV, minimax* — 2025

- Converted screen captures into board state with template matching and HSV ownership detection, then selected moves with summed-area tables and 3-ply minimax with alpha-beta pruning.

## Technical Skills

**Languages**: Python, C, C++
**Machine learning and vision**: PyTorch, torchvision, scikit-learn, pandas, NumPy, OpenCV, Matplotlib, linear probing
**Systems and tools**: Git, pytest, asynchronous HTTP APIs, MCP, CLI packaging, JSON and Markdown pipelines